

SS-RESfM: decisions to settle before submission

which RESfM baseline anchors our comparison

Background in one paragraph

We have three supervised RESfM variants, all evaluated on identical tracks, pipeline and environment. RESfM-scratch is trained by us under our uniform protocol (checkpoint selected by validation reprojection error, infinity-point margin 10^{-4} , applied identically to every arm so the outlier mechanism is the only variable). RESfM-faithful is the same training with the authors’ two released secondary values restored (checkpoint selected by classification Accuracy, margin 10^{-3}). RESfM-released are the authors’ published weights. Measured differences (translation mean, post-BA):

	MegaDepth	1DSfM	1DSfM-hard	Strecha	BlendedMVS	Olsson
RESfM-scratch (5 seeds)	0.37 ± 0.12	10.50 ± 1.45	19.16 ± 3.42	0.39 ± 0.72	0.35 ± 0.04	7.37 ± 2.63
RESfM-faithful	0.50	9.75	17.58	0.14	0.37	8.76
RESfM-released	0.21	11.12	15.29	0.20	0.33	9.10
SS-RESfM best arm	0.36–0.50	8.93	12.42	1.97	0.14	2.91

The authors’ Accuracy rule selects epoch $\sim 5.5\text{k}$; reprojection-based selection selects $\sim 16.5\text{k}$. Faithful is therefore weaker on five of six datasets: our uniform protocol helps the baseline, and every margin we report against RESfM-scratch is a lower bound.

Questions

1. Which baseline anchors the headline comparison? RESfM-scratch (stronger, cleanest control, conservative for us) or RESfM-faithful (the authors’ shipped configuration, weaker, gives us wins on five of six datasets including in-distribution parity)? Our concern with faithful-as-primary: a reviewer, or the RESfM authors themselves, can observe that we selected checkpoints for our arms by the better rule while the baseline used the worse one.
2. How do we handle the released checkpoint? It is the strongest supervised artifact in distribution (0.209 vs. our 0.36–0.50), yet it loses to us out of distribution on our tracks (1DSfM 11.12 vs. 8.93; BlendedMVS 0.33 vs. 0.14) and collapses on Olsson (9.10). Reference row plus an explicit “no SOTA in distribution” statement, or a stronger framing?
3. What is the strongest claim we are willing to defend? Candidates: (a) mechanism-isolated study, no SOTA claim (current draft); (b) “against RESfM trained from scratch under the authors’ released configuration, on identical tracks, SS-RESfM wins five of six benchmarks”; (c) something stronger.
4. Is the checkpoint-selection result a contribution in its own right? Classification accuracy is a poor proxy for reconstruction quality (epoch 5.5k vs. 16.5k; ~ 0.15 worse on MegaDepth). It also cannot be used by a label-free method at all, since it requires ground-truth outlier labels — which is part of our argument for reprojection-based selection.

5. Does the answer differ by venue? CVPR reviewers tend to expect explicit state-of-the-art positioning; an ICLR/analysis framing tolerates the scoped version. We currently plan the mechanism paper for CVPR and a separate label-ceiling/selection paper for ICLR.
6. Do we withdraw the 3DV submission? Withdrawal frees the mechanism paper for CVPR (deadline after the 3DV decision, so nothing is lost by waiting) but forfeits a likely publication. Related: are both papers worth running in parallel, or should we consolidate?

Supporting material

Full results (translation, rotation, registered cameras; five-seed bands; classical GLOMAP/COLMAP baselines; ESFM* clean-track reference) are in the companion document RESULTS_master_table.pdf.